CSC5041 Introduction to Databases

Teaching Plan — AY2026–27, Term 1

Lecturer: Prof. Lin Xuemin

Tutors: Chen Junan (Tutorials 2, 4, 6) • Zeng Ziyi (Tutorials 1, 3, 5)

Weekly schedule

Week	Date	Lecture topic	Tutorial	Tutor
1	9 Sep	Introduction to Databases: database systems, DBMS architecture, three-schema architecture, data independence, course overview	—	—
2	16 Sep	Conceptual Database Design: the Entity-Relationship model, entities, attributes, relationships, cardinality and participation constraints	—	—
3	23 Sep	The Relational Data Model and ER-to-relational mapping, integrity constraints; Relational Algebra: selection, projection, set operators, joins, division	Tutorial 1	Zeng Ziyi
4	30 Sep	SQL, Part I: SELECT–FROM–WHERE, sorting, aggregation, grouping, nested and correlated subqueries	—	—
A1	30 Sep	**Assignment 1 release**		
5	7 Oct	**National Day holiday**		
6	14 Oct	SQL, Part II: outer joins, set comparison, advanced queries; introduction to PL/pgSQL	Tutorial 2	Chen Junan
P	14 Oct	**Project release**		
7	21 Oct	Functional Dependencies: inference rules, attribute closure, candidate keys, minimal cover; Normal Forms 1NF–BCNF **In-class Test 1 (covers Weeks 1–6)**	—	—
8	28 Oct	Relational Database Design I: decompositions, lossless-join and dependency preservation	Tutorial 3	Zeng Ziyi
A1	*TBA*	**Assignment 1 due**		
9	4 Nov	Relational Database Design II: 3NF and BCNF design algorithms	—	—
A2	4 Nov	**Assignment 2 release**		
10	11 Nov	Disk, Files and Buffer Management: storage hierarchy, file organisations, buffer replacement policies	Tutorial 4	Chen Junan
11	18 Nov	Index Structures: ISAM, B^+-trees, extendible hashing	—	—
12	25 Nov	Transaction Management I: ACID properties, schedules, conflict serialisability, precedence graphs	Tutorial 5	Zeng Ziyi
P	25 Nov	**Project due**		
13	2 Dec	Transaction Management II: two-phase locking, timestamp ordering, deadlock handling, crash recovery **In-class Test 2 (covers Weeks 7–12)**	—	—
14	9 Dec	Graph Databases and Big Graph Processing (overview only); course review	Tutorial 6	Chen Junan
A2	*TBA*	**Assignment 2 due**		
Exam	*TBA*	**Final Examination**		

In-class tests are held during the lecture slot of the week shown; assignments are submitted online by the due date shown.

Tutorial overview

Tutorial	Week	Date	Tutor	Theme
1	3	23 Sep	Zeng Ziyi	Conceptual and logical database design: ER modelling and mapping to relations
2	6	14 Oct	Chen Junan	Relational algebra and basic SQL
3	8	28 Oct	Zeng Ziyi	Advanced SQL and functional dependencies
4	10	11 Nov	Chen Junan	Normalisation and relational database design
5	12	25 Nov	Zeng Ziyi	Storage, buffer management and index structures
6	14	9 Dec	Chen Junan	Transaction management

Tutorial 1 — Week 3 — 23 Sep

Conceptual and logical database design

Tutor: Zeng Ziyi **Lectures covered:** Weeks 1–3

Topics

- Entities, attributes and keys; weak entities.
- Cardinality and participation constraints.
- ER-to-relational mapping: primary keys, foreign keys, and when a relationship needs its own table.

Tutorial 2 — Week 6 — 14 Oct

Relational algebra and basic SQL

Tutor: Chen Junan **Lectures covered:** Weeks 4–6

Topics

- Relational algebra: select, project, join, set difference, division.
- SQL: `GROUP BY`, `HAVING`, `NOT EXISTS`, and `NOT IN` with NULLs.

Tutorial 3 — Week 8 — 28 Oct

Advanced SQL and functional dependencies

Tutor: Zeng Ziyi **Lectures covered:** Weeks 7–8

Topics

- Outer joins and `COUNT(DISTINCT ...)`.
- Attribute closure and candidate keys.
- Minimal cover.

Tutorial 4 — Week 10 — 11 Nov

Normalisation and relational database design

Tutor: Chen Junan **Lectures covered:** Weeks 9–10

Topics

- Candidate keys and the highest normal form.
- Minimal cover; lossless-join and dependency-preserving 3NF decomposition.
- BCNF decomposition and dependency preservation.

Tutorial 5 — Week 12 — 25 Nov

Storage, buffer management and index structures

Tutor: Zeng Ziyi **Lectures covered:** Weeks 11–12

Topics

- Buffer replacement: LRU, MRU and FIFO compared by hit rate.
- B^+-tree insertion and node splitting.

Tutorial 6 — Week 14 — 9 Dec

Transaction management

Tutor: Chen Junan **Lectures covered:** Weeks 13–14

Topics

- Conflicts and the precedence graph.
- Two-phase locking and strict 2PL.
- Recoverable, cascadeless and strict schedules.
- Recovery with write-ahead logging and checkpoints: redo and undo.
- Deadlock: detection and prevention.